

Physics Mock Test

Wave Optics

Class 12 | UNIT 6 (B)

Marks: 24

Time: 60 Minutes

Syllabus Scope: Huygen's principle, Coherent sources, Interference (YDSE), Diffraction, and Polarization (Brewster's Law).

General Instructions: Clearly state the assumptions made (e.g., small angle approximation) in your derivations.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Interference Conditions

In Young's Double-Slit Experiment, for constructive interference (Bright Fringes) to occur at a point, the path difference between the two waves must be:

- a) $n\lambda$
- b) $(n + \frac{1}{2})\lambda$
- c) $n\lambda/2$
- d) Odd multiple of $\lambda/2$

2. Diffraction Observation

The diffraction of light is most clearly noticeable when the size of the obstacle or aperture is:

- a) Much larger than λ
- b) Infinite
- c) Comparable to or smaller than λ
- d) Exactly zero

3. Nature of Light Waves

The phenomenon of polarization can only be exhibited by waves that are:

- a) Longitudinal
- b) Transverse
- c) Sound waves
- d) Stationary scalar waves

4. Fringe Patterns

The fringe width (β) in Young's double-slit experiment depends on λ , distance between slits d , and screen distance D as:

a) $\beta = \lambda D / d$

b) $\beta = d / (\lambda D)$

c) $\beta = \lambda d / D$

d) $\beta = D / (\lambda d)$

5. Single-Slit Central Maxima

The angular width of the central maximum in a single-slit diffraction pattern is given by:

a) λ / a

b) $2\lambda / a$

c) $\lambda D / a$

d) $2\lambda D / a$

6. Polarization by Reflection

Brewster's Law relates the refractive index (n) to the polarizing angle (i_p) by the equation:

a) $n = \sin(i_p)$

b) $n = \cos(i_p)$

c) $n = \tan(i_p)$

d) $n = 1/\tan(i_p)$

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. YDSE Fringe Calculation

In a YDSE setup, the slits are separated by 0.5 mm and the screen is placed 2.0 m away. If light of wavelength 600 nm is used, calculate the fringe width of the resulting interference pattern.

8. Analyzing Path Difference

Constructive interference is observed at a point where the path difference is $1.8 \mu\text{m}$. Determine the wavelength of the light if this point corresponds to the third-order bright fringe ($n=3$).

9. Diffraction Grating Maxima

A diffraction grating has 500 lines per mm. Light of wavelength 500 nm falls normally on the grating. Calculate the angular position of the second-order maximum.

10. Intensity and Polarization

Unpolarized light of intensity I_0 is passed through a single polaroid. What is the intensity of the transmitted light? State Malus' Law for the case where two polaroids are used.

11. Central Maximum and Slit Width

A single slit of width 0.1 mm is illuminated by light of wavelength 500 nm. Calculate the **linear width** of the central maximum on a screen placed 1.0 m away.

12. Thin Film Interference Logic

Explain why a thin soap film appears dark in reflected light when its thickness is very small compared to the wavelength of incident light. (Briefly mention the phase change upon reflection).

Answer Key & Solutions

1. a) $n\lambda$

Where $n = 0, 1, 2, \dots$

2. c) Comparable to λ

Diffraction is significant only when wavelength is on the same order of size as the aperture.

3. b) Transverse

Polarization restricts vibrations to a single plane; longitudinal waves oscillate only along propagation.

4. a) $\beta = \lambda D / d$

Fringe width increases with screen distance and wavelength.

5. b) $2\lambda / a$

Calculated as twice the angle from the center to the first minimum.

6. c) $n = \tan(i_p)$

At this angle, reflected and refracted rays are perpendicular.

7. 2.4 mm

$$\beta = \lambda D / d = (600 \times 10^{-9} \times 2) / (0.5 \times 10^{-3}) = 1.2 \times 10^{-6} / 0.5 \times 10^{-3} = 2.4 \times 10^{-3} \text{ m.}$$

8. 600 nm

$$\Delta p = n\lambda \rightarrow 1.8 \times 10^{-6} = 3 \times \lambda \rightarrow \lambda = 0.6 \times 10^{-6} \text{ m} = 600 \text{ nm.}$$

9. $\approx 30^\circ$

$$d = 1 / 500 \text{ mm} = 2 \times 10^{-6} \text{ m.}$$

$$\sin \theta = n\lambda / d = (2 \times 500 \times 10^{-9}) / (2 \times 10^{-6}) = 10^{-6} / 2 \times 10^{-6} = 0.5. \theta = 30^\circ.$$

10. $I_0/2$

Malus' Law: $I = I_{\text{trans}} \cdot \cos^2 \theta$, where I_{trans} is intensity from the first polaroid and θ is the angle between the two polarizers.

11. 10 mm (1.0 cm)

$$\text{Linear width } W = (2\lambda/a) \times D = (2 \times 500 \times 10^{-9} / 0.1 \times 10^{-3}) \times 1 = 10^{-2} \text{ m} = 1 \text{ cm.}$$

12. Destructive Interference

Light reflected from the upper surface (rare to dense) undergoes a π phase change,

while light from the bottom surface does not. For very thin films ($t \approx 0$), the two rays are out of phase and cancel out.

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